

LAB – Lightweight IaaS platforms

Hands on experience with Kubernetes installation

References:

- Kubernetes documentation
<https://kubernetes.io/docs/home/>

Charmed Bundle

- Charmed bundle is a type of juju installation in which everything is automated
- A Yaml file describes which components and where they should be installed
- For each component the configuration is also specified

Deploy Kubernetes (example – DO NOT RUN)

- Download a standard bundle and customize

```
wget https://github.com/charmed-kubernetes/bundle/blob/main/releases/1.27/bundle.yaml
```

- Deploy the bundle specifying where to install the controller node (node 10) and one worker (node 11)

```
juju deploy ./bundle.yaml --map-machines=existing,0=10,1=11
```

- If you have more than one worker extend the role to other machines (node 12 in this case)

```
juju add-unit --to 12 kubernetes-worker
```

Tutorial

<https://jaas.ai/kubernetes-core>

```

root@SNH0YM5GWPGME2L:~# juju deploy ./bundle.yaml
Resolving charm: cs:~containers/containerd-53
Resolving charm: cs:~containers/easyrsa-295
Resolving charm: cs:~containers/etcd-485
Resolving charm: cs:~containers/flannel-466
Resolving charm: cs:~containers/kubeapi-load-balancer-701
Resolving charm: cs:~containers/kubernetes-master-788
Resolving charm: cs:~containers/kubernetes-worker-623
Executing changes:
- upload charm cs:~containers/containerd-53 for series bionic
- deploy application containerd on bionic using cs:~containers/containerd-53
- set annotations for containerd
- upload charm cs:~containers/easyrsa-295 for series bionic
- deploy application easyrsa on bionic using cs:~containers/easyrsa-295
  added resource easyrsa
- set annotations for easyrsa
- upload charm cs:~containers/etcd-485 for series bionic
- deploy application etcd on bionic using cs:~containers/etcd-485
  added resource core
  added resource etcd
  added resource snapshot
- set annotations for etcd
- upload charm cs:~containers/flannel-466 for series bionic
- deploy application flannel on bionic using cs:~containers/flannel-466
  added resource flannel-amd64
  added resource flannel-arm64
  added resource flannel-s390x
- set annotations for flannel
- upload charm cs:~containers/kubeapi-load-balancer-701 for series bionic
- deploy application kubeapi-load-balancer on bionic using cs:~containers/kubeapi-load-balancer-701
- expose kubeapi-load-balancer
- set annotations for kubeapi-load-balancer
- upload charm cs:~containers/kubernetes-master-788 for series bionic
- deploy application kubernetes-master on bionic using cs:~containers/kubernetes-master-788
  added resource cdk-addons
  added resource core
  added resource kube-apiserver
  added resource kube-controller-manager
  added resource kube-proxy
  added resource kube-scheduler
  added resource kubectrl
- set annotations for kubernetes-master
- upload charm cs:~containers/kubernetes-worker-623 for series bionic
- deploy application kubernetes-worker on bionic using cs:~containers/kubernetes-worker-623
  added resource cni-amd64
  added resource cni-arm64
  added resource cni-s390x
  added resource core
  added resource kube-proxy
  added resource kubectrl
  added resource kubetl

```

```
root@SNHOYM5GWPGME2L:~# juju status
Model      Controller      Cloud/Region  Version  SLA          Timestamp
default    manual-controller mycloud/default 2.7.0    unsupported  14:42:27Z

App      Version  Status      Scale  Charm          Store      Rev  OS      Notes
containerd  waiting 0          0/1 containerd     jujucharms 53   ubuntu
easysrsa  waiting 1          0/1 easysrsa       jujucharms 295  ubuntu
etcd      maintenance 1          0/1 etcd           jujucharms 485  ubuntu
flannel    waiting 0          0/1 flannel        jujucharms 466  ubuntu
kubernetes-master  waiting 0/1      kubernetes-master jujucharms 788  ubuntu exposed
kubernetes-worker  maintenance 1        kubernetes-worker jujucharms 623  ubuntu exposed

Unit      Workload      Agent      Machine  Public address  Ports  Message
easysrsa/6  waiting      allocating 21/lxd/0              waiting for machine
etcd/8*    maintenance  executing  21        172.16.0.110    (install) installing charm software
kubernetes-master/7*  waiting      allocating 21        172.16.0.110    agent initializing
kubernetes-worker/8*  maintenance  executing  22        172.16.0.109    (install) installing charm software

Machine  State  DNS      Inst id      Series  AZ  Message
12       started 172.16.0.111 manual:172.16.0.111 bionic  Manually provisioned machine
21       started 172.16.0.110 manual:172.16.0.110 bionic  Manually provisioned machine
21/lxd/0 pending pending    bionic  Retrieving image: rootfs: 27% (24.50MB/s)
22       started 172.16.0.109 manual:172.16.0.109 bionic  Manually provisioned machine
```

```
root@SNHOYM5GWPGME2L:~# juju status
Model      Controller      Cloud/Region  Version  SLA          Timestamp
default    manual-controller mycloud/default 2.7.0    unsupported  15:03:22Z

App                Version  Status  Scale  Charm          Store      Rev  OS    Notes
containerd         3.0.1    active   3      containerd     juju charms 53   ubuntu
easysrsa            3.3.15   active   1      easysrsa       juju charms 295  ubuntu
etcd                0.11.0   active   1      etcd           juju charms 485  ubuntu
flannel             0.11.0   active   3      flannel        juju charms 466  ubuntu
kubernetes-master   1.17.0   active   1      kubernetes-master juju charms 788  ubuntu exposed
kubernetes-worker   1.17.0   active   2      kubernetes-worker juju charms 623  ubuntu exposed

Unit                                Workload  Agent  Machine  Public address  Ports                Message
easysrsa/6*                        active    idle    21/lxd/0  10.104.131.111
etcd/8*                             active    idle    21        172.16.0.110    2379/tcp             Healthy with 1 known peer
kubernetes-master/7*               active    idle    21        172.16.0.110    6443/tcp             Kubernetes master running.
containerd/2                      active    idle    172.16.0.110
flannel/2                          active    idle    172.16.0.110
kubernetes-worker/8*               active    idle    22        172.16.0.109    80/tcp,443/tcp       Kubernetes worker running.
containerd/1*                     active    idle    172.16.0.109
flannel/1*                         active    idle    172.16.0.109
kubernetes-worker/9                active    executing 12      172.16.0.111
containerd/3                      active    idle    172.16.0.111
flannel/3                         active    idle    172.16.0.111

Machine  State  DNS           Inst id          Series  AZ  Message
12       started 172.16.0.111  manual:172.16.0.111 bionic  Manual provisioned machine
21       started 172.16.0.110  manual:172.16.0.110 bionic  Manual provisioned machine
21/lxd/0 started 10.104.131.111 juju-6bdf2f-21-lxd-0 bionic  Container started
22       started 172.16.0.109  manual:172.16.0.109 bionic  Manual provisioned machine
```

Minikube

- An alternative installation method is available for test environment
- This method is named *minikube* in which a Kubernetes installation is run in a single environment
- Minikube exploits docker as container engine

Install Docker

- Install some pre-requisites

```
sudo apt install apt-transport-https ca-certificates curl  
software-properties-common
```

- Add the key of the official docker repository in the system

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg |  
sudo apt-key add -
```

- Add the Docker repository

```
sudo add-apt-repository "deb [arch=amd64]  
https://download.docker.com/linux/ubuntu jammy stable"
```

- Install Docker

```
sudo apt install docker-ce
```


Minikube

- Download minikube

```
curl -Lo minikube  
https://storage.googleapis.com/minikube/releases/latest/minikube-  
linux-amd64 \ && chmod +x minikube
```

```
mkdir -p /usr/local/bin/  
install minikube /usr/local/bin/
```

Install some dependencies

```
apt-get install conntrack
```

```
wget https://github.com/Mirantis/cni-dockerd/releases/download/v0.3.1/cni-dockerd\_0.3.1.3-0.ubuntu-jammy\_amd64.deb
```

```
dpkg -i cni-dockerd_0.3.1.3-0.ubuntu-jammy_amd64.deb
```

```
wget https://github.com/kubernetes-sigs/cni-tools/releases/download/v1.27.0/crictl-v1.27.0-linux-amd64.tar.gz
```

```
tar xvfz crictl-v1.27.0-linux-amd64.tar.gz
```

```
install crictl /usr/local/bin/
```

```
wget https://github.com/container networking/plugins/releases/download/v1.2.0/cni-plugins-linux-amd64-v1.2.0.tgz
```

```
mkdir -p /opt/cni/bin
```

```
tar xvfz cni-plugins-linux-amd64-v1.2.0.tgz --directory /opt/cni/bin
```

Installation

- Install the environment

```
minikube start --driver=none
```

- Check the status

```
minikube status
```

```
ubuntu@haproxy:~$ sudo minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
ubuntu@haproxy:~$ sudo minikube start --driver=none
minikube v1.10.1 on Ubuntu 18.04
Using the none driver based on user configuration
Starting control plane node minikube in cluster minikube
Running on localhost (CPUs=4, Memory=3944MB, Disk=19673MB) ...
OS release is Ubuntu 18.04.3 LTS
Preparing Kubernetes v1.18.2 on Docker 19.03.8 ...
  ■ kubelet.resolv-conf=/run/systemd/resolve/resolv.conf
  > kubectrl.sha256: 65 B / 65 B [-----] 100.00% ? p/s 0s
  > kubeadm.sha256: 65 B / 65 B [-----] 100.00% ? p/s 0s
  > kubelet.sha256: 65 B / 65 B [-----] 100.00% ? p/s 0s
  > kubectrl: 41.99 MiB / 41.99 MiB [-----] 100.00% 45.19 MiB p/s 1s
  > kubeadm: 37.97 MiB / 37.97 MiB [-----] 100.00% 25.53 MiB p/s 1s
  > kubelet: 108.03 MiB / 108.03 MiB [-----] 100.00% 59.93 MiB p/s 2s

[15] Configuring local host environment ...

? The 'none' driver is designed for experts who need to integrate with an existing VM
? Most users should use the newer 'docker' driver instead, which does not require root!
? For more information, see: https://minikube.sigs.k8s.io/docs/reference/drivers/none/

? kubectrl and minikube configuration will be stored in /home/ubuntu
? To use kubectrl or minikube commands as your own user, you may need to relocate them. For example, to overwrite your own settings, run:

  ■ sudo mv /home/ubuntu/.kube /home/ubuntu/.minikube $HOME
  ■ sudo chown -R $USER $HOME/.kube $HOME/.minikube

? This can also be done automatically by setting the env var CHANGE_MINIKUBE_NONE_USER=true
? Verifying Kubernetes components...
? Enabled addons: default-storageclass, storage-provisioner
$ Done! kubectrl is now configured to use "minikube"
? For best results, install kubectrl: https://kubernetes.io/docs/tasks/tools/install-kubectrl/
```

Interact with the cluster

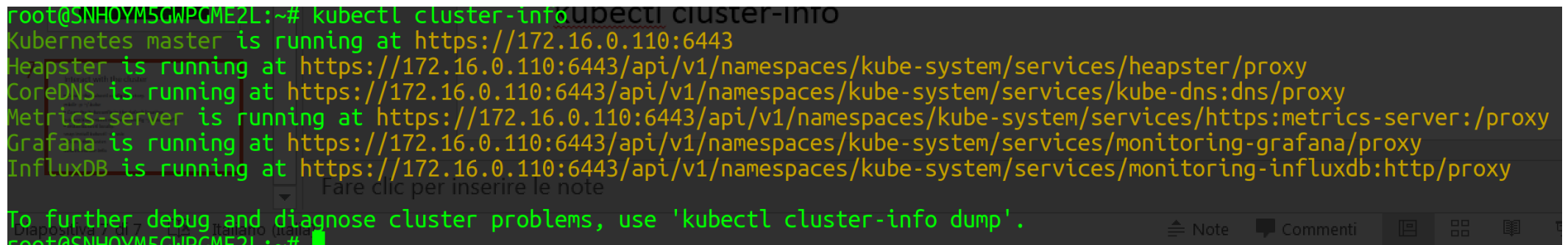
- Install kubectl

```
apt install snapd
```

```
snap install kubectl --classic
```

- Query the cluster.

```
kubectl cluster-info
```

A terminal window screenshot showing the output of the 'kubectl cluster-info' command. The output lists the Kubernetes master and various services like Heapster, CoreDNS, Metrics-server, Grafana, and InfluxDB, all running at the same URL. The terminal has a dark background with green and yellow text. At the bottom, there is a status bar with icons for notes, comments, and other functions.

```
root@SNHOYM5GWPGE2L:~# kubectl cluster-info
Kubernetes master is running at https://172.16.0.110:6443
Heapster is running at https://172.16.0.110:6443/api/v1/namespaces/kube-system/services/heapster/proxy
CoreDNS is running at https://172.16.0.110:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy
Metrics-server is running at https://172.16.0.110:6443/api/v1/namespaces/kube-system/services/https:metrics-server:/proxy
Grafana is running at https://172.16.0.110:6443/api/v1/namespaces/kube-system/services/monitoring-grafana/proxy
InfluxDB is running at https://172.16.0.110:6443/api/v1/namespaces/kube-system/services/monitoring-influxdb:http/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
root@SNHOYM5GWPGE2L:~#
```

Get nodes

```
kubectl get nodes
```

```
root@SNH0YM5GWPGME2L:~# kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
jpvhbab2n8ymtoa     Ready    <none>   26m   v1.17.0
ns0s334qagxcdw1     Ready    <none>   17m   v1.17.0
root@SNH0YM5GWPGME2L:~#
```

Activate some plugins (only for minikube)

```
minikube addons enable metrics-server
```

```
minikube addons enable metallb
```

```
root@haproxy:~# minikube addons list
```

ADDON NAME	PROFILE	STATUS
dashboard	minikube	disabled
default-storageclass	minikube	enabled ✓
efk	minikube	disabled
freshpod	minikube	disabled
gvisor	minikube	disabled
helm-tiller	minikube	disabled
ingress	minikube	disabled
ingress-dns	minikube	disabled
istio	minikube	disabled
istio-provisioner	minikube	disabled
logviewer	minikube	disabled
metallb	minikube	disabled
metrics-server	minikube	enabled ✓
nvidia-driver-installer	minikube	disabled
nvidia-gpu-device-plugin	minikube	disabled
registry	minikube	disabled
registry-aliases	minikube	disabled
registry-creds	minikube	disabled
storage-provisioner	minikube	enabled ✓
storage-provisioner-gluster	minikube	disabled

Configure the metallb

```
minikube addons configure metallb
```

```
developer@cloud2023:~$ minikube addons configure metallb
-- Enter Load Balancer Start IP: 192.168.1.50
-- Enter Load Balancer End IP: 192.168.1.100
    ■ Using image quay.io/metallb/speaker:v0.9.6
    ■ Using image quay.io/metallb/controller:v0.9.6
    ✓ metallb was successfully configured
developer@cloud2023:~$ █
```